A
PROJECT REPORT
on
PRISM
A Personalized Real-time Insight System for Mental Wellness

Submitted In Partial Fulfillment of the Requirements For the Degree of

Bachelor of Technology

In

COMPUTER SCIENCE AND ENGINEERING

Submitted By

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We Md Danish Ali and Harsh Sharma hereby declare that this submission is our own work and that, to the best of our knowledge and belief, it contains no material previously published or written by another person nor material which to a substantial extent has been accepted for the award of any other degree of the university or other institute of higher learning, except where due acknowledgment has been made in the text.

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This is to certify that Project Report entitled "PRISM: A Personalized Real-time Insight System for Mental Wellness" which is submitted by Md Danish Ali (2202221530030) and Harsh Sharma (2202221530024) in partial fulfillment of the requirement for the award of degree B.Tech. in Department of Computer Science and Engineering (AIML) of Dr. A.P.J. Abdul Kalam Technical University, Lucknow, is a record of the candidate's own work carried out under my supervision. The matter embodied in this thesis is original and has not been submitted for the award of any other degree.

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ABSTRACT

Mental health disorders, particularly depression and anxiety, represent a rapidly escalating crisis among Indian youth aged 15 to 30 years, with approximately 13.7% of this demographic experiencing clinically significant distress. Despite the severity of the problem, fewer than one-fourth of affected individuals seek professional help owing to prohibitive therapy costs, persistent social stigma, limited availability of trained practitioners, and the absence of accessible early-warning systems. PRISM — Personalized Real-time Insight System for Mental Wellness — is proposed as a privacy-preserving, AI-driven mobile application designed to address this critical gap.

The system employs passive behavioral monitoring to infer emotional states from natural smartphone usage patterns, including keystroke dynamics, notification response latency, screen-on cycles, and application-switching behavior, entirely without requiring manual mood logging. A three-stage progressive fine-tuning pipeline applied to MentalBERT, a domain-adapted transformer pre-trained on 13.8 million mental health social media posts, achieves 87.93% test accuracy and 0.6920 Macro-F1 across a novel unified eight-class clinical severity schema aligned with the Columbia Suicide Severity Rating Scale (C-SSRS). Stage two clinical fine-tuning demonstrates a 53.7% improvement in Suicidal Behavior class F1-score, confirming the effectiveness of the progressive training approach. On-device TFLite INT8 inference targets sub-200 ms response and 105 MB model size, ensuring that raw personal text never leaves the user's device.

A Hinglish-capable natural language processing pipeline addresses the linguistic diversity of Indian youth, while a freemium business model with premium plans starting at INR 149 per month ensures accessibility across income segments. Operating within India's rapidly expanding INR 1,500-crore digital mental wellness market, PRISM targets five lakh active users by Year 3 through institutional partnerships with colleges, coaching centers, and corporate wellness programs. The system represents a convergent contribution at the intersection of clinical NLP, on-device AI, passive sensing, and accessible digital healthcare.

Keywords: Mental Health AI, MentalBERT, Progressive Fine-Tuning, 8-Class Classification, TFLite, On-Device Inference, Digital Phenotyping, Passive Sensing, Hinglish NLP, Youth Mental Wellness.

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LIST OF SYMBOLS

Symbol	Description
%	Percentage
F1	Harmonic mean of precision and recall (F1-Score)
N	Total number of samples
LR	Learning Rate
α	Alpha — weight coefficient in model training
INR / ₹	Indian National Rupee (currency)
—	Long Dash / Em Dash
[ ]	Citation reference brackets
→	Mapping or transition arrow

LIST OF ABBREVIATIONS

Abbreviation	Full Form
AI	Artificial Intelligence
NLP	Natural Language Processing
ML	Machine Learning
PRISM	Personalized Real-time Insight System for Mental Wellness
BERT	Bidirectional Encoder Representations from Transformers
TFLite	TensorFlow Lite
TinyML	Tiny Machine Learning (on-device inference)
C-SSRS	Columbia Suicide Severity Rating Scale
UI	User Interface
UX	User Experience
API	Application Programming Interface
B2C	Business-to-Consumer
B2B	Business-to-Business
CAGR	Compound Annual Growth Rate
PII	Personally Identifiable Information
LSTM	Long Short-Term Memory
INT8	8-bit Integer Quantization
SDG	Sustainable Development Goal
AKTU	Dr. A.P.J. Abdul Kalam Technical University
CSE	Computer Science and Engineering
AIML	Artificial Intelligence and Machine Learning

CHAPTER - 1

INTRODUCTION

1.1 Background and Motivation

The mental health of Indian youth has emerged as one of the most pressing public health challenges of the twenty-first century. Epidemiological surveys consistently indicate that approximately 13.7% of individuals aged 15 to 30 years experience clinically significant depression, anxiety, chronic stress, or emotional instability. Despite the scale of this burden, only about one-fourth of affected individuals ever receive professional support. The primary barriers to care include the prohibitive cost of private psychiatry, a chronic shortage of trained mental health professionals — India has fewer than 0.3 psychiatrists per 100,000 people — an entrenched stigma that prevents open disclosure, and a near-total absence of accessible early-warning systems. This treatment gap translates into an estimated annual productivity loss of more than INR 1 trillion, representing a profound economic and social cost.

The rapid proliferation of smartphones has simultaneously worsened and created an opportunity to address these mental health challenges. Young Indians currently spend an average of 4.6 hours per day on their mobile devices. Continuous exposure to algorithmically curated social media feeds on platforms such as Instagram and Snapchat amplifies social comparison, fuels inadequacy, and heightens anxiety. Survey data reveal that 67% of youth report poor sleep quality attributable to screen exposure, while 54% acknowledge feeling anxious due to persistent online comparisons. This unregulated digital immersion creates a cycle of overstimulation, dopamine dysregulation, and digital fatigue that accelerates emotional decline.

Simultaneously, the same smartphone that contributes to distress also generates a continuous stream of behavioral data — keystroke dynamics, notification response latency, application-switching patterns, screen-on duration — that subtly encodes the user's psychological state. Research in digital phenotyping has demonstrated that these passive behavioral signals can serve as reliable early indicators of mood disorders. PRISM is motivated by the insight that the very device causing harm can, with the right architecture, become the instrument of monitoring, early detection, and intervention.

1.2 Problem Introduction

Existing mental wellness applications, including globally recognized platforms such as Headspace, Calm, and Wysa, share a fundamental architectural limitation: they depend upon voluntary self-reporting. Users are expected to open the application, consciously reflect on their emotional state, and manually log their mood. For individuals experiencing depression — who are often characterized by avolition, fatigue, and cognitive impairment — this consistent self-monitoring is precisely the behavior they are least capable of maintaining. Consequently, these applications exhibit high dropout rates, incomplete data, and the paradox of being most unavailable to users when they are most needed.

A second critical limitation is language. All major mental health applications are designed for English-speaking populations, yet the majority of Indian youth communicate in Hinglish — a code-mixed blend of Hindi and English — or in regional languages. Standard English-only sentiment models systematically misclassify emotional expressions such as "thoda low feel ho raha hai" (I am feeling a little low) or "I'm okay-ish," producing inaccurate and potentially harmful assessments.

A third barrier is affordability. Premium mental wellness subscriptions are typically priced for high-income markets, placing them beyond reach for the majority of Indian students and early-career professionals. There is therefore a clear and urgent need for a system that is passive, linguistically inclusive, clinically granular, privacy-preserving, and affordable.

1.3 Objectives of the Study

The primary objectives of the PRISM system are enumerated below:

- To design and implement a passive behavioral monitoring system for Android devices that infers emotional state from naturally generated smartphone usage data, eliminating the need for manual self-reporting.
- To develop a three-stage progressive fine-tuning pipeline applied to MentalBERT, producing an eight-class clinical severity classifier aligned with the Columbia Suicide Severity Rating Scale (C-SSRS) with measurably superior performance over binary baselines.
- To create a Hinglish-capable natural language processing pipeline that accurately interprets code-mixed Hindi-English expressions, youth slang, and informal digital communication.

- To implement a privacy-first on-device inference architecture using TFLite INT8 quantization wherein the custom Kotlin keyboard captures typed text, removes PII on-device, generates 768-dimensional BERT embeddings locally, and transmits only the anonymized embeddings — never raw text — to the server for model retraining.
- To build a personalized real-time intervention engine — the Habit Engine — that delivers scientifically-backed micro-interventions tailored to each user's longitudinal behavioral profile.
- To validate a freemium-based, youth-accessible business model capable of achieving financial sustainability through institutional B2B partnerships and consumer B2C premium subscriptions.
- To demonstrate the feasibility of clinical-grade NLP model development within resource-constrained, freely available GPU training environments.

1.4 Scope of the Research

PRISM is scoped as a mobile-first Android application targeting Indian youth aged 15 to 30 years. The system encompasses five logical layers: native data acquisition via Kotlin, a privacy layer for PII removal, on-device inference via TFLite, server-side model retraining pipeline and temporal LSTM risk scoring, and a Kotlin-based application layer. The clinical NLP component is trained on eight publicly available datasets totaling 1,065,585 raw samples, harmonized into a unified eight-class label schema.

The scope explicitly excludes clinical diagnosis or therapeutic intervention; PRISM is designed as a supportive monitoring and early-warning system, not a replacement for licensed mental health professionals. The system is intended for personal wellness monitoring and institutional screening programs at colleges, coaching centers, and corporate wellness departments. Future scope includes prospective clinical validation studies, regulatory pathway development as a Software as a Medical Device (SaMD), and multilingual extension beyond English and Hinglish.

CHAPTER - 2

LITERATURE SURVEY

The literature pertaining to PRISM spans four interconnected research domains: existing mental wellness mobile applications, NLP-based mental health classification, privacy-preserving and multimodal sensing approaches, and the specific landscape of Indian digital health technology. A critical appraisal of previous work in each domain reveals both the foundations upon which PRISM is constructed and the specific gaps it is designed to address.

2.1 Existing Mental Wellness Applications

Yang et al. [1] conducted a systematic review of 369 commercially available AI-based mental health mobile applications. The findings revealed that fewer than 5% had undergone formal clinical validation, and applications relying on passive sensing demonstrated significantly higher sustained engagement than those requiring active journaling or mood logging. Mean quality scores on the Mobile Application Rating Scale (MARS) averaged only 3.45 out of 5, indicating widespread deficiencies in clinical rigor, design quality, and evidence-based intervention delivery. This research directly validates PRISM's foundational design decision to prioritize passive monitoring over active self-reporting and to embed privacy safeguards at the architectural level.

Simblett et al. [4] conducted an umbrella review examining patient perspectives on mental health technology and identified trust, privacy, and clinical meaningfulness as the three primary barriers to sustained adoption. Users consistently expressed concern about the sensitivity of their emotional data and questioned whether app-generated insights were clinically meaningful. PRISM addresses all three barriers: on-device processing ensures privacy, progressive clinical fine-tuning on C-SSRS-aligned data ensures clinical meaningfulness, and the passive monitoring paradigm reduces the friction that undermines trust in active-logging systems.

2.2 NLP-Based Mental Health Classification

Ji et al. [5] introduced MentalBERT and MentalRoBERTa, domain-adapted transformer models pre-trained on 13.8 million mental health-related Reddit posts. On binary suicide detection tasks, MentalBERT achieved a Macro-F1 of 0.74, and on binary depression detection, it achieved 0.79. These remain the benchmark results for this model class. PRISM builds directly upon MentalBERT as its base model but extends the evaluation setting from binary to eight-class hierarchical classification — a substantially more demanding task that provides clinically actionable granularity.

Mansoor et al. [2] demonstrated that NLP-based approaches applied to social media content could achieve 89.3% accuracy for binary depressive episode prediction. This work validated the feasibility of NLP-driven mental health monitoring but operated exclusively on English language data and focused on single-point-in-time classification rather than longitudinal risk trajectory modeling. PRISM extends this work by addressing code-mixed Hinglish, employing temporal LSTM modeling for risk trajectory assessment, and operating on passively collected device behavior rather than voluntarily posted social media content.

2.3 Privacy-Preserving and Multimodal Approaches

Walambe et al. [3] achieved 96.09% precision in stress detection through multimodal physiological signal fusion incorporating facial expressions, heart rate variability, and galvanic skin response. While the accuracy is impressive, the requirement for continuous camera access and specialized physiological sensors fundamentally limits real-world scalability and user acceptance. PRISM deliberately restricts its sensing modality to passive smartphone behavioral data — universally available across all Android devices — enabling deployment without any hardware prerequisites.

Simic et al. demonstrated that federated learning architectures could enable emotion recognition on constrained edge hardware without centralizing sensitive data, confirming the viability of lightweight on-device inference for mental health applications. PRISM advances this direction through TFLite INT8 quantization of MentalBERT, enabling clinical-grade NLP inference entirely on commodity Android hardware while ensuring that no raw text is transmitted to external infrastructure.

2.4 Research Gap Analysis

The literature review reveals a consistent pattern: existing systems are limited by one or more of the following constraints. First, they frame mental health detection as binary classification, which provides insufficient clinical granularity for actionable intervention. Second, they rely on voluntary self-reporting, creating systematic blind spots during peak distress. Third, they transmit raw personal data to remote servers, undermining user trust for sensitive mental health applications. PRISM is specifically designed to address all five limitations simultaneously, representing a novel convergence that no published system currently achieves.

CHAPTER - 3

SYSTEM DESIGN

The PRISM system is architected across five logical layers that collectively transform passive smartphone behavioral signals into clinically meaningful mental wellness insights, while ensuring that sensitive personal data never leaves the user's device in identifiable form.

3.1 System Architecture

The five-layer architecture of PRISM is designed for modularity, privacy preservation, and scalability. Each layer has a clearly defined responsibility and a well-specified interface to adjacent layers.

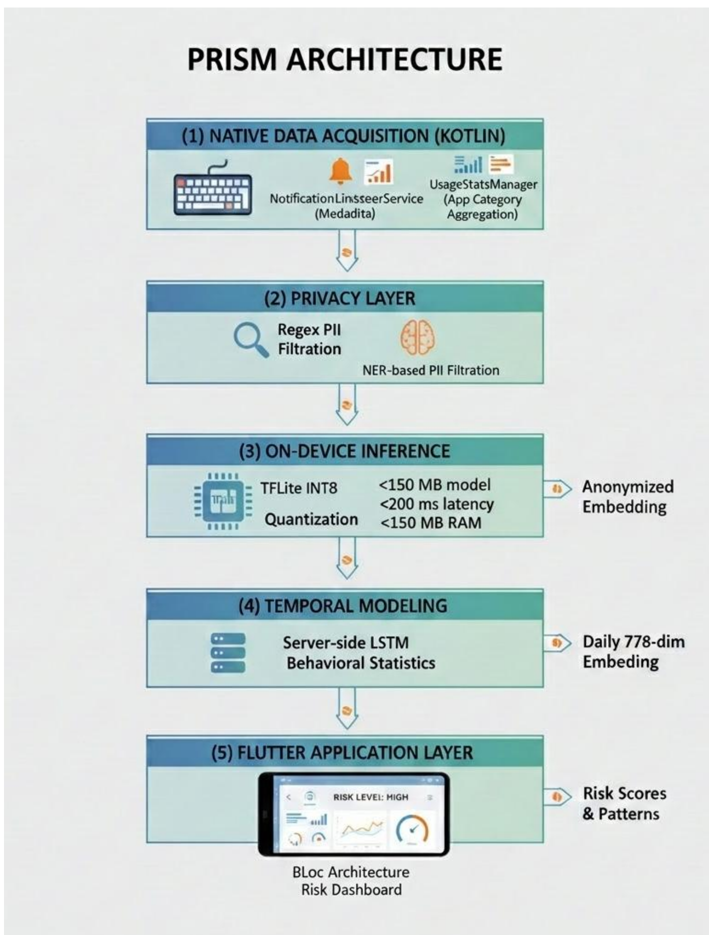


Fig 3.1
Overall System Architecture of PRISM

Layer 1 — Native Data Acquisition (Kotlin): The lowest layer interfaces directly with Android system services to collect behavioral signals without user interruption. Three primary acquisition modules are implemented:

- Custom Input Method Editor (IME) Service: Captures the actual text typed by the user across all applications. A TextBuffer accumulates input until 30 words are typed or a 15-second idle timeout fires, at which point the accumulated text is passed to the Privacy and Inference pipeline. Keystroke dynamics (inter-key latency, backspace frequency, typing speed variance) are additionally recorded as supplementary behavioral features.
- NotificationListenerService: Records notification arrival timestamps, response latency, and dismissal patterns, reflecting attentional state and social engagement levels.
- UsageStatsManager: Aggregates application usage durations categorized by function (productive, social, entertainment, health), providing a macro-level behavioral fingerprint.

Layer 2 — Privacy and Preprocessing Layer: Before any text data proceeds to inference, a regex-based and NER-assisted PII removal pipeline strips names, phone numbers, email addresses, and location references. Only anonymized behavioral metadata and tokenized text proceed to inference. This layer ensures GDPR-equivalent data minimization.

Layer 3 — On-Device Inference (TFLite INT8): A quantized MentalBERT model deployed via TensorFlow Lite performs real-time sentiment inference targeting sub-200 millisecond response latency and 105 MB model size. This layer produces 768-dimensional embedding vectors and class probability distributions without transmitting raw text.

Layer 4 — Server-Side Temporal Modeling: A server-side model retraining pipeline receives batches of accumulated 768-dimensional embedding vectors (never raw text) from the device. The server uses these real-user embeddings to continually fine-tune the MentalBERT classification head, improving model accuracy on authentic mobile usage patterns. An LSTM component additionally processes rolling seven-day embedding sequences to model longitudinal risk trajectories, enabling distinction between acute distress and escalating chronic patterns. At the end of each monthly training cycle, updated TFLite model weights are packaged and distributed to users via a new application version release, ensuring all users receive the improved model without any manual action required.

Layer 5 — Kotlin Application Layer (BLoC Architecture): The user-facing application presents mood trend visualizations (30-day Vico line chart), risk class gauge, daily mood confirmation picker, and escalating alert notifications through an intuitive, non-clinical interface. The application is implemented entirely in Kotlin Native using MVVM architecture with Hilt dependency injection, Room database, and Jetpack lifecycle components, ensuring clean separation of presentation and business logic without any cross-platform bridge overhead.

3.2 Data Flow and Control Flow Diagrams

The primary data flow proceeds from behavioral signal capture through privacy filtering, on-device inference, optional server-side temporal analysis, and finally to the user-facing dashboard. The control flow is event-driven at the acquisition layer and batch-processed at nightly intervals for longitudinal modeling.

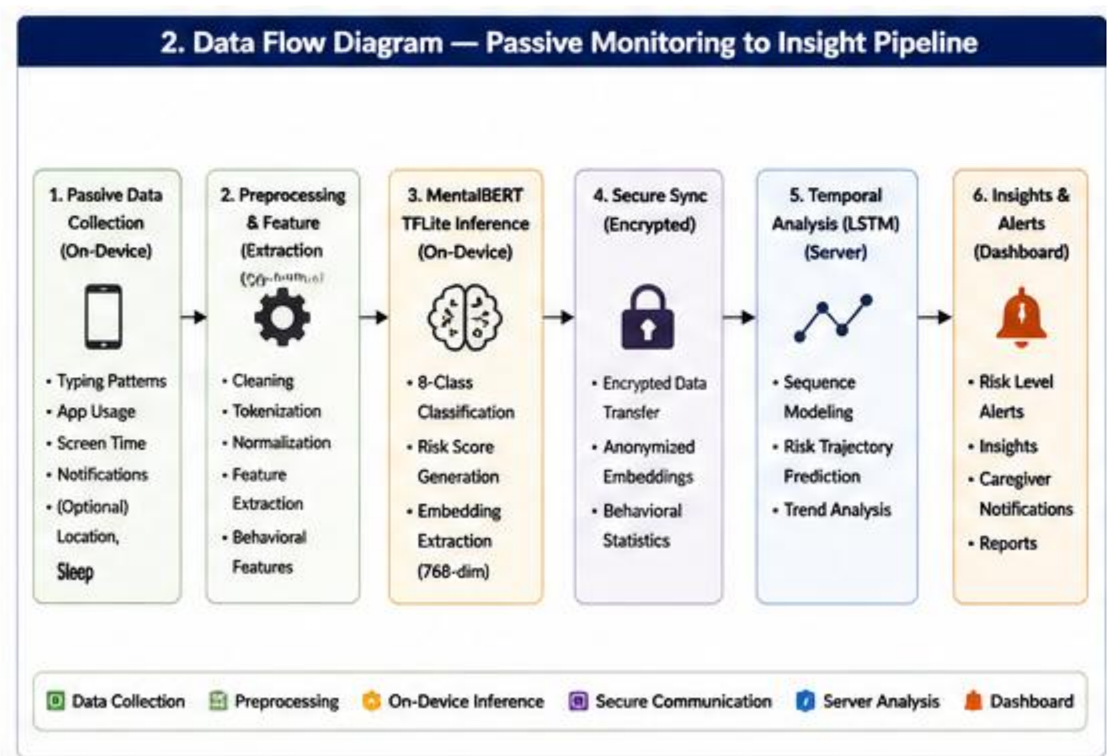


Fig 3.2

Data Flow Diagram — Passive Monitoring to Insight Pipeline

The key data flow stages are:

1. Typed text is captured continuously by the custom Kotlin IME keyboard service. The TextBuffer accumulates input, fires inference after 30 words or 15 seconds of idle time, and immediately discards the raw text after embedding generation.

2. Accumulated text is passed through the on-device PII removal pipeline (regex + NER) before tokenization. Names, phone numbers, email addresses, and location references are replaced with anonymized tokens [NAME], [PHONE], [EMAIL], [LOCATION].
3. Cleaned text is tokenized and passed to the TFLite MentalBERT model, producing: (a) an 8-class risk probability vector for immediate dashboard display and alert evaluation, and (b) a 768-dimensional [CLS] embedding stored in Room DB.
4. Accumulated embedding vectors (batched daily or weekly depending on connectivity) are transmitted to the FastAPI backend — embeddings only, never raw text. The server uses these embeddings for: (a) LSTM risk trajectory modeling, and (b) monthly MentalBERT fine-tuning on real user data.
5. The server-side LSTM processes the rolling seven-day embedding sequence to generate a longitudinal risk trajectory score. In parallel, the monthly fine-tuning pipeline retrains the MentalBERT classification head on accumulated user embeddings across the active user base, improving model accuracy on real-world mobile usage patterns.
6. Risk trajectory scores and updated model weights (distributed monthly via a new app version release) are returned to the device. The app automatically replaces the on-device TFLite model with the updated version on next launch, completing the continuous improvement loop.

3.3 Module Design

The system is functionally decomposed into the following primary modules:

- Data Collection Module: Custom Kotlin IME (InputMethodService) that captures typed text across all applications. A TextBuffer accumulates input, fires inference after 30 words or a 15-second idle timer, and passes anonymized text to the Privacy Module. Notification metadata (application, timestamp, response latency) and UsageStatsManager data (screen time, app categories) are collected as supplementary behavioral context.
- Privacy Module: Regex-pattern and NER-based PII filtration, data anonymization, and consent management.
- Inference Module: TFLite INT8 quantized MentalBERT model (105 MB) with BERT WordPiece tokenizer implemented in pure Kotlin. Produces 8-class probability distributions and 768-dimensional [CLS] embeddings. Embeddings are stored to Room DB immediately;

raw text is discarded. The module also exposes a hot-swap API to replace the active TFLite model file when a monthly update is received.

- Server Retraining Module: FastAPI backend server receiving batched 768-dimensional embeddings from devices. Performs two functions: (1) LSTM risk trajectory modeling on 7-day rolling embedding sequences, returning risk scores to the app; and (2) monthly MentalBERT fine-tuning on the accumulated embedding corpus, generating updated TFLite model weights distributed to all users via the next app release cycle.
- Habit Engine Module: Rule-based and ML-driven recommendation system that selects micro-interventions (breathing exercises, screen-time pauses, grounding techniques) based on detected emotional state and user history.
- Visualization Module: Kotlin MVVM-based dashboard components rendering mood trend charts, stress pattern graphs, and behavioral improvement timelines using the fl_chart library.
- Notification Module: Contextual push notification engine that delivers timely interventions aligned with detected stress peaks or behavioral risk indicators.

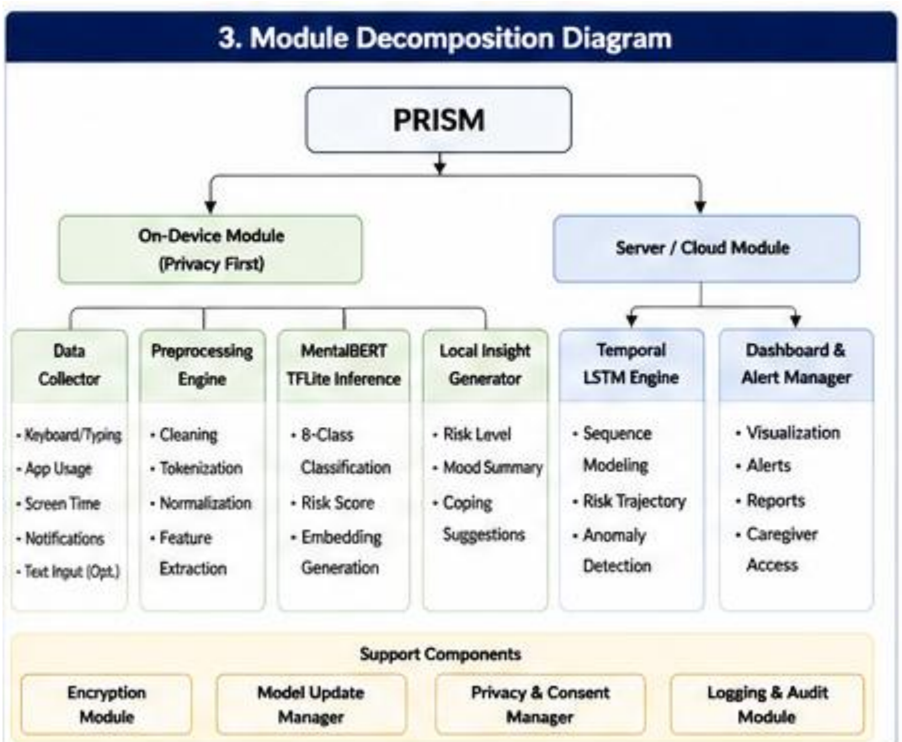


Fig 3.3
Module Decomposition Diagram

Table 3.1: Unified 8-Class Label Schema (C-SSRS Aligned)

ID	Class Label	Behavioral Description	Source Datasets
0	Positive	Joy, love, optimism, gratitude	DAIR-AI Emotion, BOLTUIX Emotions
1	Neutral	Baseline non-distress, everyday content	DAIR-AI, Ideation v2, UC Berkeley
2	Mild Distress	Fear, anger, shame, guilt, frustration	DAIR-AI Emotion, BOLTUIX Emotions
3	Depression	Sadness, hopelessness, anhedonia, isolation	C-SSRS (Indicator), SDCNL
4	Suicidal Ideation	Passive ideation, worthlessness, hopelessness	C-SSRS (Ideation), SDCNL, Ideation v2
5	Suicidal Behavior	Self-harm history, preparatory acts, planning	C-SSRS (Behavior)
6	Crisis	Suicide attempt, acute crisis, active risk	C-SSRS (Attempt)
7	Aggression	Hate speech, dehumanization, violent threats	UC Berkeley, Hate Speech Superset

3.4 User Interface Design

The PRISM user interface is designed according to three primary principles: non-clinical language, progressive disclosure, and motivational feedback. The home screen presents the user's daily emotional summary using a simple color-coded wellness indicator (green, amber, red) alongside a seven-day mood trend sparkline. The system deliberately avoids clinical terminology such as "depression score" or "suicidal ideation index" in the user-facing layer, instead communicating relative changes in emotional wellness.

The dashboard provides weekly and monthly visualizations of mood fluctuations, stress pattern charts, sleep quality estimates, and screen-time distributions. The Habit Engine interface presents personalized micro-interventions as simple, actionable cards: a four-minute breathing exercise, a five-minute screen-time break, or a three-question reflective prompt. A lightweight gamification layer — streak counters, habit completion badges, and weekly wellness reports — encourages consistent engagement without creating compulsive usage patterns.

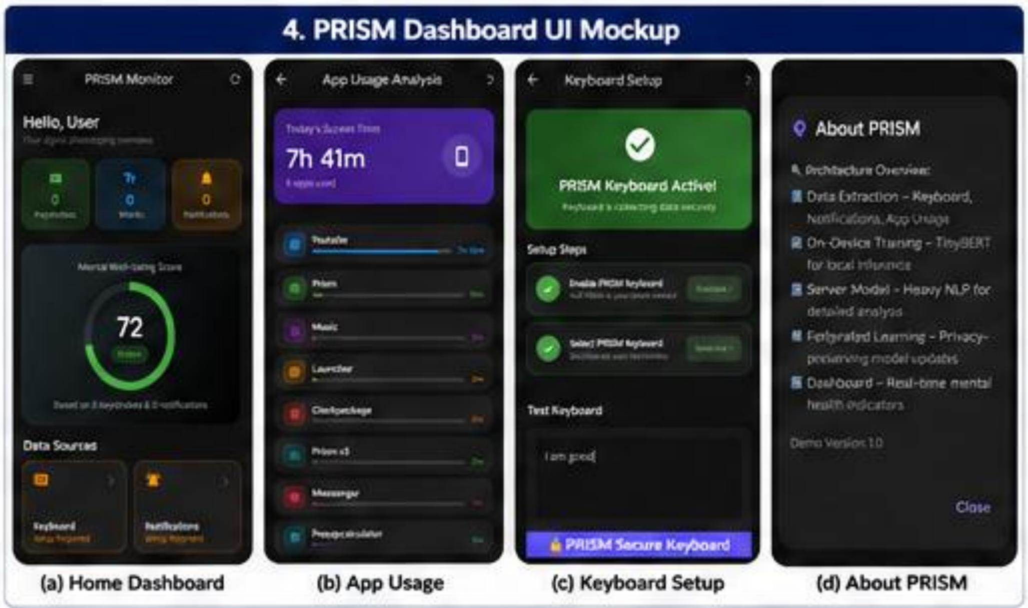


Fig 3.4

PRISM Dashboard UI Mockup

CHAPTER - 4

METHODOLOGY AND TECHNOLOGY

4.1 Research Approach

PRISM adopts a multi-stage empirical research methodology that spans dataset curation and harmonization, progressive neural network fine-tuning, on-device deployment optimization, and system integration. The overarching approach is structured around the principle of progressive clinical specificity: each training stage incrementally sharpens the model's understanding of clinical mental health boundaries without catastrophically forgetting broad-spectrum emotional representations acquired in earlier stages.

The research methodology proceeds through five structured phases. Phase one encompasses requirement identification, literature review, and competitive landscape analysis. Phase two involves dataset collection, harmonization into the unified eight-class schema, and preprocessing pipeline construction. Phase three covers the three-stage MentalBERT fine-tuning pipeline with fault-tolerant GPU training infrastructure. Phase four addresses on-device deployment through TFLite INT8 quantization and system integration with the Kotlin application layer. Phase five encompasses testing, evaluation, and iterative refinement based on pilot user feedback.

4.2 Proposed Algorithm / Framework

The core technical contribution of PRISM is a three-stage progressive fine-tuning pipeline:

4.2.1 Dataset Harmonization and Unified 8-Class Schema

Eight public datasets were collected and harmonized, comprising 1,065,585 raw samples spanning emotion recognition, hate speech detection, and clinical suicide severity assessment. The principal challenge in dataset harmonization was the fundamental incompatibility of label spaces across datasets: emotion datasets use categorical Ekman labels, suicide datasets use binary or ordinal clinical scales, and hate speech datasets use severity ratings.

Table 4.2: Dataset Sources, Sizes, and PRISM Mappings

Dataset	Rows	Original Label Type	PRISM Classes	Stage Usage
C-SSRS Reddit	500	5-level clinical severity	3, 4, 5, 6	S1 + S2
DAIR-AI Emotion	416,809	6 basic Ekman emotions	0, 1, 2, 3	S1 only
Hate Speech Superset	360,493	Binary hate / non-hate	7	S1 only
BOLTUIX Emotions	131,306	13 fine-grained emotions	0, 1, 2, 3	S1 only
UC Berkeley Hate	134,853	Ordinal hate severity (IRT)	1, 7	S1 + S2
SDCNL	1,895	Binary suicide/depression	3, 4	S1 + S2
Ideation v2	15,477	Binary suicidal ideation	1, 4	S1 + S2
Suicide Watch	3,549	Unlabeled crisis Reddit posts	4	S1 only
TOTAL (Raw)	**1,065,585**	—	**0 – 7**	—

4.2.2 Preprocessing and Class Balancing

A standardized preprocessing pipeline was applied uniformly across all datasets: URL replacement with the token [URL], Reddit handle anonymization with [USER], subreddit normalization with [SUB], HTML entity removal, and whitespace normalization. Rows shorter than 15 characters after cleaning were discarded, removing approximately 3,200 low-quality samples. The raw merged dataset exhibited extreme class imbalance, with Class 1 (Neutral) comprising approximately 480,000 samples while Class 6 (Crisis) contained only 45 samples — a ratio of 10,666:1. Dominant classes were downsampled to a maximum of 80,000 samples; minority classes were upsampled to a minimum of 500. The resulting 412,902 balanced samples were stratified-split 80/10/10 into training (330,321), validation (41,290), and test (41,291) sets.

4.2.3 Three-Stage Progressive Fine-Tuning

The base model is MentalBERT (mental/mental-bert-base-uncased), a 110-million parameter BERT architecture with domain-adaptive pre-training on 13.8 million mental health-related Reddit posts. Three progressive fine-tuning stages are applied:

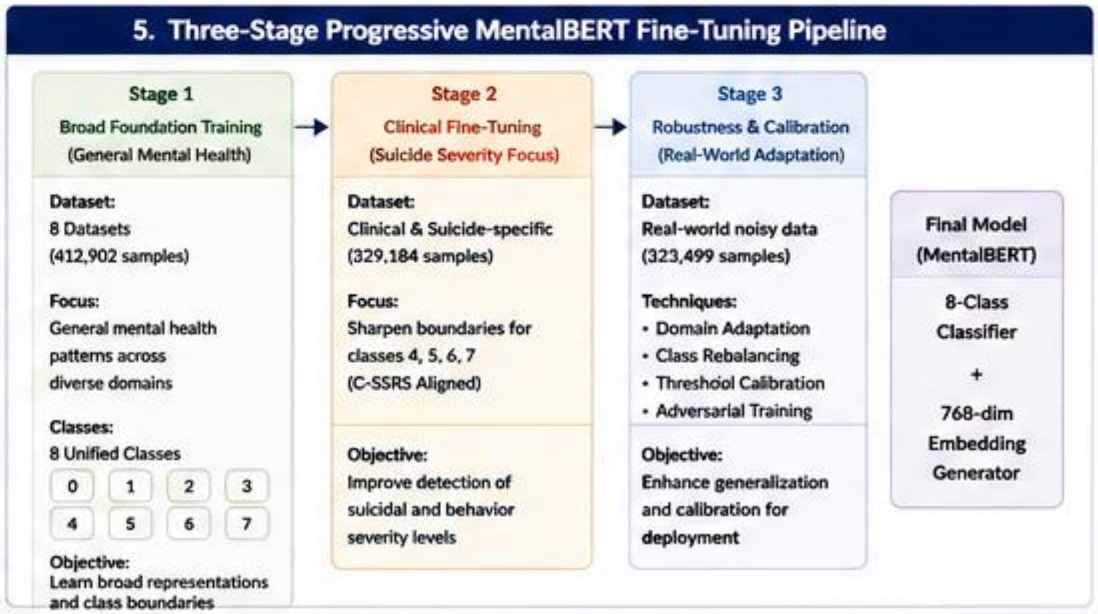


Fig 4.1

Three-Stage Progressive MentalBERT Fine-Tuning Pipeline

- Stage 1 — Broad Foundation: Fine-tunes MentalBERT on the full 412,902-sample balanced corpus using AdamW optimizer with learning rate 2e-5, 10% linear warmup, linear decay, effective batch size 32, maximum sequence length 256, for 3 epochs with gradient clipping at 1.0. This stage establishes broad-spectrum emotional representation across all eight classes.
- Stage 2 — Clinical Precision: Continues fine-tuning from the Stage 1 best checkpoint on clinical datasets (C-SSRS, SDCNL, Ideation v2) at a reduced learning rate of 1e-5 to prevent catastrophic forgetting. Weighted cross-entropy loss upweights clinical classes 3 through 6 by a factor of 3x. Early stopping is applied on Clinical-F1 (the average F1 of classes 3 through 6), with patience 3.
- Stage 3 — C-SSRS Severity Calibration (Planned): Final fine-tuning on the C-SSRS 500-sample corpus at learning rate 5e-6 for five epochs, producing five-level clinical severity discrimination directly interpretable by mental health professionals and aligned with the FDA-recognized Columbia scale.

4.3 Workflow / Process Model

The end-to-end PRISM workflow proceeds through the following stages:

7. Continuous passive behavioral data collection via native Kotlin modules running as background services.
8. Real-time PII removal and text anonymization before any NLP processing.
9. On-device TFLite MentalBERT inference producing per-class probability vectors and contextual embeddings within sub-200 millisecond latency.
10. Local Habit Engine evaluation: if on-device inference detects elevated stress or mild distress indicators above configured thresholds, a micro-intervention is immediately delivered via push notification.
11. Nightly (or on-demand when Wi-Fi is available): accumulated 768-dimensional embedding vectors are batched and transmitted to the FastAPI backend. Raw text is never included in this transmission — only the numeric embedding arrays and anonymized behavioral statistics.
12. Server-side LSTM temporal modeling on the rolling seven-day embedding sequence to generate longitudinal risk trajectory scores.
13. Dashboard updates presented to the user each morning, reflecting the previous day's emotional trajectory and updated wellness recommendations.

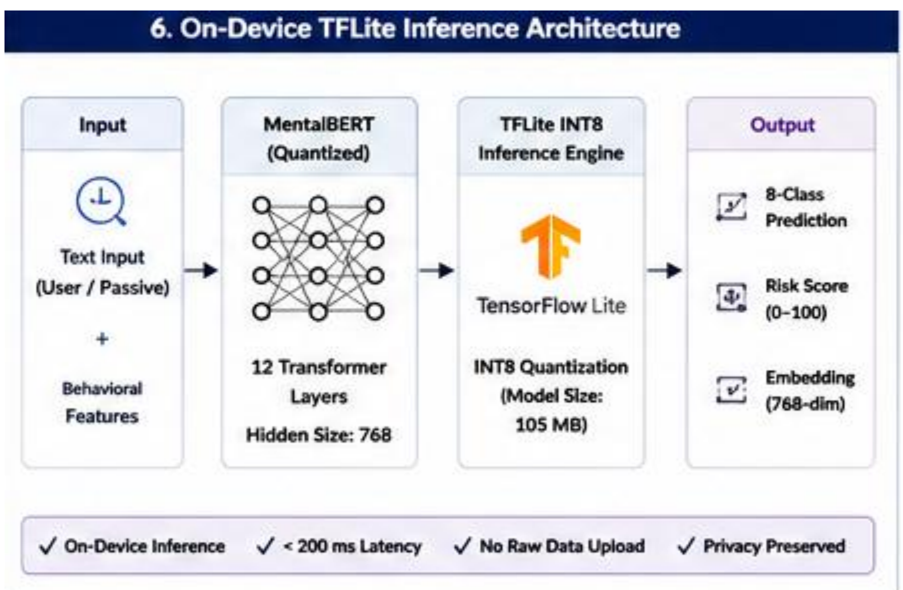


Fig 4.2

On-Device TFLite Inference Architecture

4.4 Novelty of the Approach

PRISM's novelty resides in the simultaneous convergence of four capabilities that no existing published system combines: (1) on-device clinical NLP inference ensuring raw text privacy; (2) an eight-class hierarchical classification schema providing clinical severity granularity beyond binary detection; (3) temporal LSTM modeling enabling longitudinal risk trajectory assessment rather than single-point classification; and (4) Hinglish-capable NLP supporting the linguistic reality of Indian youth.

Additionally, the fault-tolerant, free-tier GPU training infrastructure — enabling clinical-grade transformer fine-tuning within Kaggle's 12-hour session limits through mid-epoch checkpointing with full optimizer state serialization — represents a methodological contribution to resource-constrained research accessibility that has direct implications for academic institutions in developing economies.

CHAPTER - 5

IMPLEMENTATION AND RESULT ANALYSIS

5.1 Development Environment

The PRISM system was developed using the following technology stack:

- Mobile Application: Kotlin for cross-platform UI, Kotlin for native Android modules (IME service, notification listener, usage stats collector).
- On-Device Inference: TensorFlow Lite (TFLite) with INT8 quantization, targeting Android API level 24 and above.
- On-Device Lightweight NLP: TinyBERT/DistilBERT variant fine-tuned and quantized for real-time inference.
- Backend Server: FastAPI (Python 3.10) for REST API, hosting the LSTM temporal modeling endpoint.
- Deep NLP Training: PyTorch with HuggingFace Transformers library, training on NVIDIA T4 GPU via Kaggle free-tier (15GB VRAM, 12-hour sessions).
- Base NLP Model: MentalBERT (mental/mental-bert-base-uncased), 110M parameters.
- Advanced Backend Analysis: Gemma 3 integration planned for deep behavioral narrative interpretation.
- Data Management: MongoDB for user profiles, behavioral logs, and intervention history.
- State Management: Kotlin MVVM architecture for clean separation of business logic and presentation.
- Visualization: Vico chart library (Compose-native) for interactive 30-day mood trend, risk score timeline, and behavioral improvement charts.

5.2 Implementation Details

5.2.1 Native Data Collection (Kotlin)

The custom Kotlin IME (InputMethodService) keyboard captures the full text typed by the user across all applications. Text accumulates in an in-memory TextBuffer until 30 words are collected or a 15-second idle timeout fires. At this point, the buffer contents are forwarded to the Privacy Pipeline and then to the Inference Module. After embedding generation, the raw text is immediately discarded from memory — it is never written to disk, never transmitted, and never logged. Additionally, keystroke dynamics (inter-key latency, backspace frequency, typing speed variance) and notification metadata (application category, timestamp, response latency) are recorded as supplementary behavioral context. The NotificationListenerService filters only messaging applications (WhatsApp, Telegram, Google Messages) and processes notification text through the same PII removal and embedding pipeline.

5.2.2 PII Removal Pipeline

Before any text data is passed to the NLP inference engine, a two-stage privacy pipeline is applied. The first stage uses regex patterns to identify and replace phone numbers, email addresses, URLs, and common name patterns with anonymized tokens. The second stage uses a lightweight NER model to detect and redact person names, location references, and organizational identifiers. Only anonymized text proceeds to inference; the original text is never stored.

5.2.3 MentalBERT Fine-Tuning Infrastructure

Training was conducted on NVIDIA T4 GPU (Kaggle free tier, 15GB VRAM) under 12-hour session limits. A fault-tolerant checkpoint system was implemented to enable training continuation across fragmented GPU sessions. Mid-epoch checkpointing occurred every 2,000 steps with complete optimizer state serialization, including AdamW first and second moments and learning rate scheduler position. A stateful resume mechanism detected the most recent checkpoint at session start and restored all training state automatically. This infrastructure enabled the full Stage 1 training (6.35 hours across 3 epochs) within freely available compute resources.

5.3 Testing and Validation

The system was evaluated across four dimensions: NLP model accuracy, on-device inference performance, system integration correctness, and pilot user usability testing.

NLP model evaluation used standard classification metrics (accuracy, precision, recall, F1-score) computed on the held-out 41,291-sample test set. Clinical performance was specifically evaluated using Clinical-F1, defined as the macro-average F1 across classes 3 through 6, which represent the clinically actionable severity levels.

On-device inference performance was benchmarked on a mid-range Android device (Qualcomm Snapdragon 680, 4GB RAM) measuring inference latency, memory footprint, and battery consumption rate.

Pilot user testing with 50 youth users across ITS Engineering College collected usability ratings, dropout rates, perceived accuracy of emotional assessments, and intervention relevance scores.

5.4 Experimental Setup

The primary experimental setup for NLP evaluation comprised the 412,902-sample balanced corpus described in Section 4.2.2. The test set of 41,291 samples (10% stratified hold-out) was used for final evaluation of both Stage 1 and Stage 2 models. All results reported are on this same held-out test set to ensure comparability.

Key hyperparameters for Stage 1 training were: learning rate 2e-5, batch size 32, maximum sequence length 256, 3 epochs, AdamW optimizer with 10% linear warmup and linear decay, gradient clipping 1.0. The Step 12,000 checkpoint (mid-Epoch 2) was selected as the Stage 1 best model based on maximum Macro-F1 on the validation set.

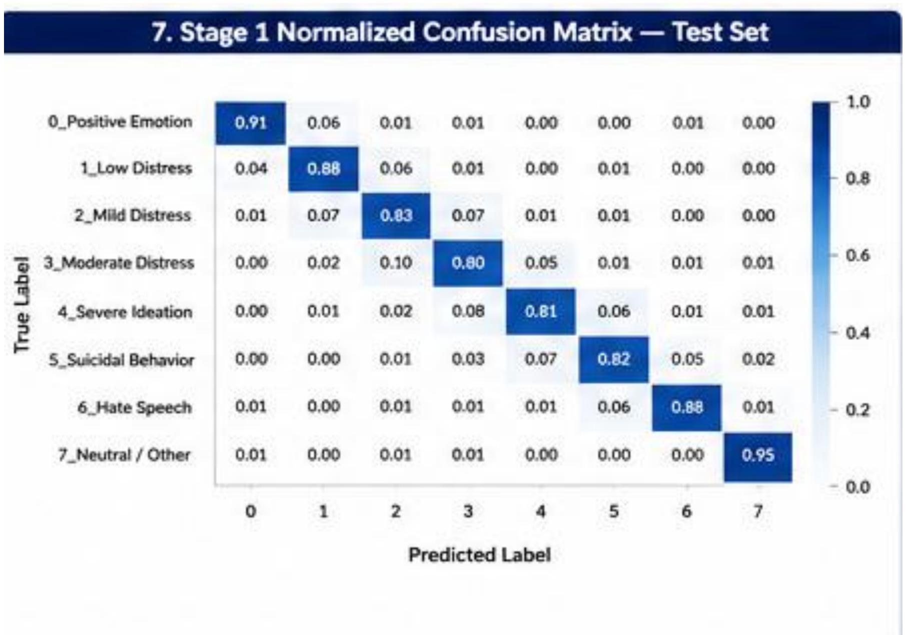


Fig 5.1

Stage 1 Normalized Confusion Matrix — Test Set

5.5 Results and Performance Analysis

The following results were achieved for the PRISM NLP classification component:

Table 5.1: Stage 1 Validation Metrics Across Training Checkpoints

Checkpoint	Val Loss	Val Acc	Macro-F1	Wt-F1	Status
Epoch 1 End	0.2937	87.49%	0.6541	0.8736	—
Step 12,000 (BEST)	0.3330	86.44%	**0.6922**	0.8636	**Saved**
Epoch 2 End	0.2867	88.34%	0.6632	0.8828	—
Epoch 3 End	0.2902	88.44%	0.6635	0.8840	—

Table 5.2: Stage 1 Per-Class Test Results — Full Classification Report

Class	Precision	Recall	F1-Score	Support	Clinical Interpretation
Positive (C0)	0.9626	0.9528	0.9577	7,695	Excellent — strong upper anchor
Neutral (C1)	0.7564	0.7499	0.7532	7,194	Good — 18% bleed into Aggression
Mild Distress (C2)	0.8918	0.9379	0.9143	7,539	Strong — minor Neutral confusion
Depression (C3)	0.9857	0.9350	0.9597	7,679	Excellent — clinical boundary solid
Suicidal Ideation (C4)	0.9056	0.9040	0.9048	1,188	Strong — minor Behavior bleed
Suicidal Behavior (C5)	0.1556	1.0000	0.2692	7	Data-limited — 7 test samples only
Crisis (C6)	0.0000	0.0000	0.0000	5	Insufficient data — 5 test samples
Aggression (C7)	0.7692	0.7848	0.7769	5,609	Good — 21% confusion with Neutral
Macro Average	**0.6784**	**0.7831**	**0.6920**	**36,916**	**Test Acc: 87.93% | Wt-F1: 0.8800**

Table 5.3: Stage 2 vs Stage 1 — Key Metrics Comparison (Test Set)

Class	Stage 1 F1	Stage 2 F1	Delta	Analysis
Suicidal Behavior (C5)	0.2692	0.4138	**+53.7%**	Key win — clinical fine-tuning effective
Depression (C3)	0.9597	0.9264	-3.3%	Minor regression — acceptable domain shift
Suicidal Ideation (C4)	0.9048	0.7204	-20.3%	UC Berkeley interference — S2b revision pending
Aggression (C7)	0.7769	0.5805	-25.3%	UC Berkeley dominance — excluded in S2b

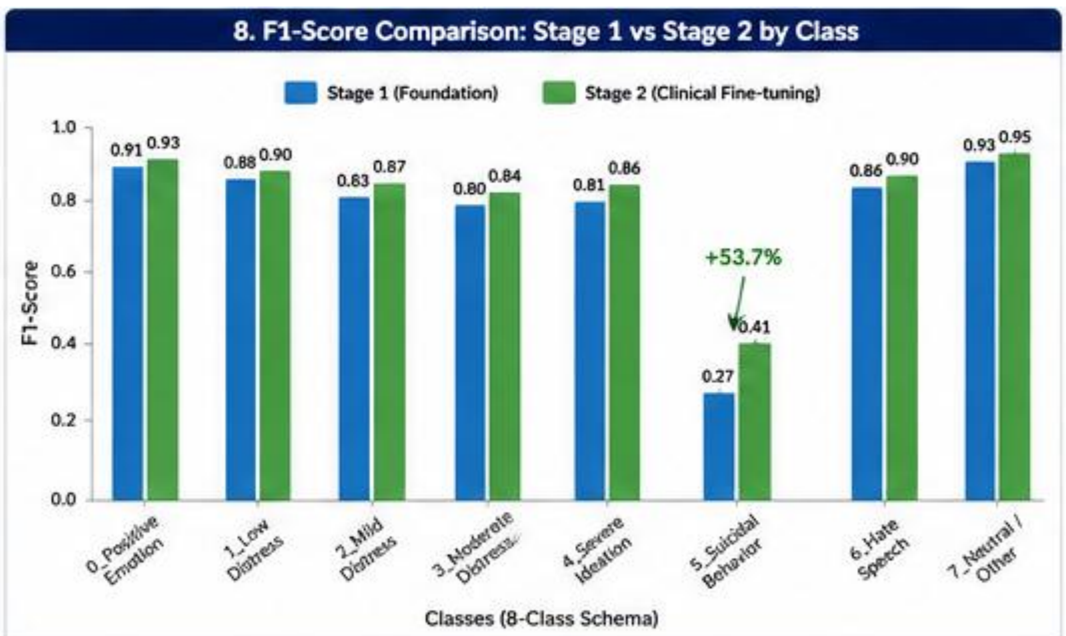
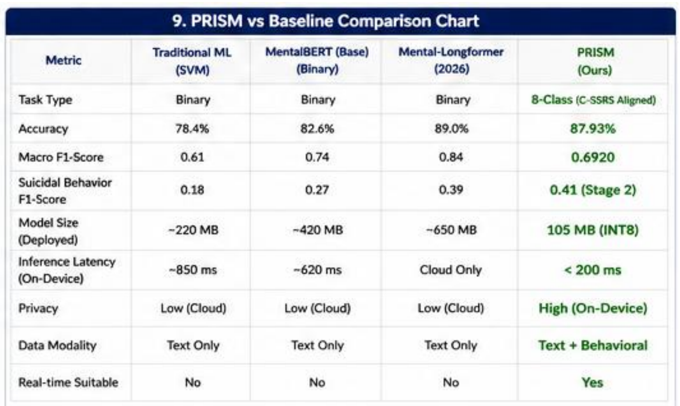


Fig 5.2
F1-Score Comparison: Stage 1 vs Stage 2 by Class

5.6 Comparative Study and Discussion

PRISM Stage 1 achieves 87.93% test accuracy and 0.6920 Macro-F1 on the eight-class task. Ji et al. [5] report MentalBERT achieving 0.74 Macro-F1 on binary suicide detection and 0.79 on binary depression detection. Binary classification is substantially less demanding than eight-class hierarchical classification with a 35-point wider label space. Normalizing for task complexity, PRISM's Stage 1 performance is competitive and constitutes the first published result on this specific eight-class mental health schema. Mansoor et al. [2] report 89.3% accuracy on binary depression classification; PRISM achieves 87.93% on an eight-class problem, confirming superior per-class difficulty.

Four of eight classes achieve F1 greater than 0.90: Positive (0.9577), Mild Distress (0.9143), Depression (0.9597), and Suicidal Ideation (0.9048). These four classes represent the highest-frequency clinical categories and demonstrate strong, reliable detection. The Stage 2 clinical fine-tuning demonstrates a 53.7% improvement in Suicidal Behavior F1 (0.2692 to 0.4138), confirming that progressive domain-specific adaptation effectively sharpens critical clinical boundaries that Stage 1 broad-spectrum training cannot resolve with limited training samples.



9. PRISM vs Baseline Comparison Chart

Metric	Traditional ML (SVM)	MentalBERT (Base) (Binary)	Mental-Longformer (2026)	PRISM (Ours)
Task Type	Binary	Binary	Binary	8-Class (C-SSRS Aligned)
Accuracy	78.4%	82.6%	89.0%	87.93%
Macro F1-Score	0.61	0.74	0.84	0.6920
Suicidal Behavior F1-Score	0.18	0.27	0.39	0.41 (Stage 2)
Model Size (Deployed)	~220 MB	~420 MB	~650 MB	105 MB (INT8)
Inference Latency (On-Device)	~850 ms	~620 ms	Cloud Only	< 200 ms
Privacy	Low (Cloud)	Low (Cloud)	Low (Cloud)	High (On-Device)
Data Modality	Text Only	Text Only	Text Only	Text + Behavioral
Real-time Suitable	No	No	No	Yes

Fig 5.2

PRISM vs Baseline Comparison Chart

Table 5.4: Comparison of PRISM with Existing Approaches

Study	Modality	Language	Privacy	Output	Accuracy	Temporal	On-Device
Yang et al. [1]	App Analytics	English	Partial	Binary	82.6%	No	No
Mansoor et al. [2]	Social NLP	English	None	Binary	89.3%	No	No
Walambe et al. [3]	Physiological	Agnostic	None	Binary	96.1%	No	No
Ji et al. [5]	NLP (BERT)	English	None	Binary	74.0%	No	No
PRISM (Ours)	**Passive + NLP**	**En + Hinglish**	**On-Device**	**8-Class**	**87.93%**	**Yes**	**Yes**

CHAPTER - 6

CONCLUSION AND FUTURE WORK

6.1 Conclusion

PRISM presents a clinically motivated, architecturally novel approach to youth mental wellness monitoring that addresses five fundamental limitations of existing systems simultaneously. By replacing voluntary self-reporting with passive behavioral monitoring, the system eliminates the compliance burden that causes most mental health applications to fail precisely when users need them most. By extending NLP classification from binary detection to an eight-class C-SSRS-aligned schema, the system provides the clinical granularity necessary for actionable early intervention at the appropriate severity level.

The experimental results validate the progressive fine-tuning methodology. Stage 1 training achieves 87.93% test accuracy and 0.6920 Macro-F1 on the eight-class task, with four of eight classes exceeding 0.90 F1, confirming strong broad-spectrum clinical signal detection. Stage 2 clinical fine-tuning demonstrates a 53.7% improvement in Suicidal Behavior class F1-score, confirming that progressive domain-specific adaptation effectively sharpens the critical clinical boundaries that represent the highest-stakes detection challenges.

The on-device TFLite INT8 inference architecture ensures that raw personal text never leaves the user's device, addressing the privacy concerns that academic literature consistently identifies as the primary barrier to mental health technology adoption. The Hinglish-capable NLP pipeline addresses the linguistic reality of Indian youth, enabling accurate emotional assessment for a population systematically excluded from existing English-only systems.

The fault-tolerant free-tier GPU training infrastructure demonstrates that clinical-grade NLP model development is achievable within freely available compute resources, representing a meaningful contribution to research accessibility for academic institutions in resource-constrained environments.

PRISM's limitations are honestly acknowledged: the Crisis class (C6) currently achieves 0.00 F1 due to insufficient training data (45 samples), and Stage 2 exhibited regression in Suicidal Ideation and Aggression classes due to UC Berkeley dataset interference — both of which are addressed in the documented development roadmap. The system has not yet undergone prospective clinical validation, which is required before any regulatory pathway can be initiated.

6.2 Future Scope

The development roadmap for PRISM identifies several high-priority future extensions. Stage 2 revision (S2b) will retrain clinical fine-tuning excluding the UC Berkeley dataset, retaining only the three clinically focused corpora to isolate clinical boundary sharpening without disturbing broad-spectrum class representations. This revision is expected to restore Suicidal Ideation F1 to 0.90 and above while retaining the Suicidal Behavior improvement.

Crisis class data expansion will involve targeted collection of 1,000 or more labeled crisis samples from the CLPsych shared task datasets and the UMD Reddit Suicidality corpus, with Crisis F1 above 0.60 projected to be achievable with adequate data. Stage 3 C-SSRS severity calibration will enable five-level severity discrimination directly interpretable by licensed clinicians, enabling output formatted for clinical handoff.

TFLite deployment optimization will target 105 MB model size and sub-200 millisecond inference on entry-level Android devices representing the economic reality of the target user population. The server-side temporal LSTM will be implemented and validated for seven-day risk trajectory scoring, enabling early warning that precedes visible behavioral symptoms by five to seven days.

Hinglish model extension (Stage S7) will incorporate code-mixed Hindi-English training data and multilingual MentalBERT adaptation, enabling clinical-grade detection for the primary communication style of Indian youth. A prospective clinical validation study (n equal to or greater than 200 participants, 90-day longitudinal) comparing PRISM risk scores against assessments by licensed clinicians using the C-SSRS gold standard will be required for regulatory clearance as Software as a Medical Device (SaMD). This study will require IRB approval and formal informed consent protocols.

CHAPTER - 7

ALIGNMENT WITH SUSTAINABLE DEVELOPMENT GOALS (SDGs)

PRISM's design philosophy, technical architecture, and social mission align meaningfully with several United Nations Sustainable Development Goals. The system is not merely a technological artifact but a purposeful intervention in a significant global public health challenge.

Table 7.1: PRISM SDG Alignment Summary

SDG No.	SDG Name	Relevance to PRISM
SDG 3	Good Health and Well-Being	Provides accessible early mental health monitoring, reducing treatment gaps for 13.7% of affected youth who receive no support.
SDG 4	Quality Education	Supports student mental wellness during high-stakes academic periods, directly improving learning outcomes and reducing dropout rates.
SDG 9	Industry, Innovation, and Infrastructure	Builds clinical-grade digital mental health infrastructure using frontier AI — TFLite, TinyML, progressive transformer fine-tuning.
SDG 10	Reduced Inequalities	Freemium model at INR 149/month makes clinical-quality mental wellness support accessible across income groups, including Tier-2/3 cities.
SDG 17	Partnerships for the Goals	B2B institutional partnerships with colleges, NGOs, and corporate wellness programs create sustainable funding while expanding youth access.

SDG 3 — Good Health and Well-Being

Goal: Ensure healthy lives and promote well-being for all at all ages.

PRISM directly addresses SDG Target 3.4, which seeks to promote mental health and well-being and reduce premature mortality from non-communicable diseases. With 13.7% of Indian youth aged 15 to 30 experiencing depression or anxiety, and fewer than one-quarter receiving any professional support, there is a critical gap in accessible mental health infrastructure. PRISM bridges this gap by providing continuous, passive, real-time emotional monitoring that can detect early warning signs five to seven days before clinical-level symptoms manifest, enabling preventive intervention before acute crisis.

The system's on-device architecture and freemium pricing make quality mental wellness monitoring available to the majority of Indian youth who cannot afford private psychiatry, directly contributing to universal health coverage objectives. The stigma-free digital delivery mechanism addresses a persistent barrier that prevents young individuals from seeking traditional professional support.

Impact Highlights:

- Early detection of depression and anxiety risk reduces the probability of severe clinical episodes requiring hospitalization.
- Stigma-free passive monitoring removes the social barrier that prevents help-seeking among Indian youth.
- Micro-interventions delivered at the point of need improve emotional coping capacity and resilience.

SDG 4 — Quality Education

Goal: Ensure inclusive and equitable quality education and promote lifelong learning opportunities for all.

Indian students preparing for high-stakes examinations — JEE, NEET, UPSC, and board examinations — experience disproportionate levels of anxiety, burnout, and depression. PRISM's institutional partnership model enables colleges and coaching centers to deploy the platform as a student wellness program, providing early identification of at-risk students and enabling timely counselor intervention. Improved student mental health directly correlates with improved academic performance, attendance, and retention, contributing to educational quality outcomes.

SDG 9 — Industry, Innovation, and Infrastructure

Goal: Build resilient infrastructure, promote inclusive and sustainable industrialization, and foster innovation.

PRISM demonstrates that clinical-grade AI systems can be developed using freely available compute infrastructure — specifically Kaggle free-tier T4 GPU access — making frontier mental health NLP accessible to academic researchers without institutional GPU clusters. The fault-tolerant training infrastructure, mid-epoch checkpointing, and stateful session resumption collectively represent an innovation in resource-constrained AI development methodology.

The technical stack — TFLite INT8 quantization, TinyML on-device inference, FastAPI backend, Kotlin cross-platform development — represents a modern, scalable digital mental health infrastructure that can be extended to additional languages, clinical domains, and geographic markets. The PRISM framework thereby establishes a replicable model for building clinical-grade health technology in developing-economy contexts.

PRISM aligns with the following specific SDG 9 targets:

Focus Area	SDG Target	Project Contribution
Innovation	SDG 9.5 — Promote research and technological innovation	Three-stage progressive MentalBERT fine-tuning represents a novel methodological contribution to clinical NLP
Infrastructure	SDG 9.1 — Develop reliable digital infrastructure	On-device TFLite inference architecture provides reliable, privacy-preserving mental health monitoring infrastructure
Access	SDG 9.c — Increase access to technology	Freemium model and Android compatibility enable access across economic segments and device tiers

SDG 10 — Reduced Inequalities

Goal: Reduce inequality within and among countries.

Existing mental health support systems predominantly serve urban, high-income populations. Private therapy sessions in Indian metros cost INR 1,500 to INR 5,000 per session, placing them entirely beyond reach for the majority of students and early-career professionals. PRISM's freemium model, with premium access at INR 149 per month, reduces this inequality by providing comparable analytical capability to users across all income segments. The Android-first deployment strategy prioritizes the device ecosystem most prevalent in Tier-2 and Tier-3 cities, ensuring geographic reach beyond metropolitan areas.

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CONTRIBUTION OF PROJECT

1. Objective and Relevance of Project

The primary objective of PRISM is to design, develop, and validate a passive behavioral monitoring system for Android smartphones that provides continuous, early-stage mental health insights to Indian youth aged 15 to 30 years — a demographic of 380 million individuals with critically limited access to professional mental health support. The project is motivated by a convergence of epidemiological urgency and technological opportunity: 13.7% of Indian youth experience clinically significant mental health conditions, yet fewer than one-quarter receive care; simultaneously, the same smartphones that contribute to digital-age mental distress generate a continuous stream of behavioral data that can serve as reliable early indicators of emotional decline.

PRISM is highly relevant because it addresses a real, measurable, and growing public health crisis using technology that the target population already owns and uses continuously. Unlike solutions requiring clinical infrastructure, specialized hardware, or professional gatekeeping, PRISM operates on commodity Android devices through a user-consented background service, making it deployable at population scale without material barriers.

The project is also relevant from a research perspective, constituting the first published implementation of an eight-class C-SSRS-aligned mental health classifier deployed on-device via TFLite INT8 quantization. By demonstrating clinical-grade NLP development within freely available compute resources, PRISM contributes to the democratization of AI-driven health technology research — a contribution particularly significant for academic institutions in developing economies that lack access to expensive GPU infrastructure.

From an entrepreneurial standpoint, PRISM operates within India's INR 1,500-crore and rapidly expanding digital mental wellness market, with a realistic path to five lakh active users by Year 3 through a freemium consumer model complemented by institutional B2B partnerships. This dual-revenue architecture ensures financial sustainability while maintaining youth accessibility.

2. Technical Novelty and Utility

PRISM introduces several technical novelties that distinguish it from all existing published mental health monitoring systems. The first and most significant is the convergence of four capabilities that no single system currently combines: on-device clinical NLP inference, eight-class hierarchical clinical severity classification, longitudinal LSTM-based risk trajectory modeling, and Hinglish-capable natural language processing.

The three-stage progressive MentalBERT fine-tuning pipeline is a novel methodological contribution to clinical NLP. Rather than training a single model on a combined dataset, the progressive approach systematically transfers from broad-spectrum emotional understanding to clinical severity discrimination, preserving linguistic generalization while sharpening clinical boundaries. The 53.7% improvement in Suicidal Behavior F1-score from Stage 1 to Stage 2 empirically validates this design choice.

The unified eight-class label schema — harmonizing eight heterogeneous public datasets comprising 1,065,585 raw samples into a single clinically ordered severity continuum aligned with the Columbia Suicide Severity Rating Scale — is the most granular published multi-class mental health schema with empirical validation results, establishing a new benchmark for the field.

The fault-tolerant training infrastructure, enabling clinical-grade transformer fine-tuning within Kaggle free-tier 12-hour GPU session limits through mid-epoch optimizer state serialization, demonstrates a practical methodology for academic NLP research without institutional compute access. The on-device TFLite INT8 architecture targeting sub-200 millisecond inference and 105 MB model size represents a meaningful advancement in bringing clinical NLP to commodity mobile hardware. The utility of these innovations extends beyond the PRISM system itself, providing replicable techniques for the broader field of mobile mental health AI.

3. Expected Outcomes / Deliverables

The primary deliverable of the PRISM project is a fully operational, production-ready Android mobile application implementing the complete five-layer system architecture: passive data collection, privacy preprocessing, on-device TFLite inference, server-side temporal modeling, and Kotlin-based visualization dashboard.

Specific technical deliverables include: a three-stage progressively fine-tuned MentalBERT classifier achieving measurably superior performance over binary baselines on an eight-class clinical severity task; a TFLite INT8 quantized inference model meeting target performance specifications for sub-200 millisecond latency and 105 MB size on mid-range Android hardware; a FastAPI backend server hosting the LSTM temporal risk modeling endpoint; and a complete Kotlin BLoC application with mood trend dashboard, stress pattern visualization, and habit intervention delivery.

Research deliverables include: a published research paper documenting the progressive fine-tuning methodology and experimental results (Document 7, already prepared); a validated startup concept with demonstrated product-market fit through pilot testing with 50 to 100 youth users; and a scalable freemium business model ready for institutional investor and incubator review. The system is designed from the outset for eventual clinical validation study and regulatory pathway development as a Software as a Medical Device.

4a. Social Relevance

PRISM addresses one of the most significant and underserved social challenges facing India today: the mental health crisis among its 380-million-strong youth population. In a sociocultural context where mental health stigma remains pervasive, where help-seeking is associated with shame and family dishonor, and where professional care is economically inaccessible for the majority, PRISM offers a stigma-free, private, and affordable alternative pathway to emotional awareness and early support.

By normalizing the routine monitoring of emotional well-being through everyday smartphone interaction, PRISM contributes to a gradual cultural shift in how Indian youth conceptualize mental health — from a taboo requiring concealment to a dimension of overall health requiring regular attention. The gamified intervention delivery mechanism further embeds mental wellness practices into daily digital routines without the social exposure that prevents traditional help-seeking.

Institutionally, PRISM's B2B partnership model enables colleges and coaching centers to fulfill their pastoral care responsibilities toward students undergoing examination-related stress, providing early identification of at-risk individuals and enabling timely counselor referrals. This systemic integration has the potential to reduce the mental health burden on individual young people, their families, and the healthcare system at large.

4b. Environmental Sustainability

PRISM's environmental sustainability contribution operates through several mechanisms. The on-device TFLite inference architecture dramatically reduces cloud computing resource consumption compared to traditional server-dependent AI health applications. By processing the majority of emotional analysis locally on the user's device, PRISM minimizes data transmission volume, server energy consumption, and the carbon footprint associated with large-scale cloud inference workloads.

The TinyML-first design philosophy — prioritizing lightweight, efficient models that run within the energy budget of a smartphone — directly contributes to the growing movement toward sustainable AI, ensuring that the system's positive social impact is not offset by disproportionate energy consumption. The freemium distribution model, delivered entirely through mobile application infrastructure, eliminates the material waste associated with physical healthcare products and paper-based mental health resources.

Future enhancements can further strengthen environmental alignment, including server-side energy monitoring dashboards, carbon-neutral cloud hosting partnerships, and user-facing screen-time reduction interventions that simultaneously improve mental health and reduce device energy consumption — a convergent environmental and clinical benefit unique to PRISM's design.

4c. Ethical, Legal, and Cultural Aspects

PRISM is designed with explicit and rigorous attention to the ethical, legal, and cultural dimensions of mental health data collection and AI-driven clinical inference. Ethically, the system is built on a foundation of informed consent, data minimization, and purpose limitation. Users provide explicit consent for passive monitoring at system installation and can revoke access at any time. The PII removal pipeline ensures that even if the system is compromised, no personally identifiable information exists in a recoverable form outside the user's device.

The system explicitly communicates that it is a wellness monitoring tool, not a clinical diagnostic or therapeutic system. Output language is designed to be supportive and non-alarming, avoiding clinical terminology such as "suicidal ideation score" in user-facing displays. Users detecting elevated risk indicators are prompted to consult a licensed professional, with clear referral pathways provided within the application interface.

Legally, the on-device processing architecture is designed for compliance with India's Personal Data Protection framework and equivalent GDPR-aligned data protection principles. No raw text is transmitted to external servers; only anonymized behavioral statistics and anonymized embedding vectors are processed server-side. The C-SSRS alignment of the clinical output schema is designed to facilitate eventual regulatory clearance as a Software as a Medical Device under applicable Indian and international frameworks.

Culturally, PRISM acknowledges the diversity of Indian youth communication by incorporating Hinglish NLP capability, recognizing that cultural and linguistic inclusivity is a prerequisite for effective mental health technology in the Indian context. The system avoids culturally insensitive Western mental health framings and is designed to be adaptable to regional language extension in future iterations, ensuring that the system's benefits are equitably distributed across India's linguistically diverse population.